

Factorising

Worked Solutions with Area-Model Diagrams

Wisest Maths — Year 1 A-Level, Algebra (ref a2)

This document contains all 50 *Factorising* questions, each with a fully worked solution and a TikZ *area-model* (box-method) diagram. Factorising is expanding run *in reverse*: the terms of the expression are placed *inside* the grid, and the factors are read off the *edges*. For a common factor the grid is a single row (the factor down the side, the bracket along the top); for a difference of two squares it is a 2×2 grid in which the two off-diagonal terms cancel; for grouping the four given terms are exactly the four interior cells.

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Factorising 1 (a2-001)

Difficulty: Foundation / Marks: 2

Factorise: $14m + 21n$

Worked solution.

The highest common factor of 14 and 21 is 7.

$$7(2m + 3n)$$

Look for highest common factors (HCF).

Final answer: $7(2m + 3n)$

Area-model diagram.

×	$2m$	$3n$
7	$14m$	$21n$

HCF 7 down the side; read the other factor $(2m + 3n)$ across the top.

Factorising 2 (a2-002)

Difficulty: Foundation / Marks: 2

Factorise: $5x^3y^2 - 15x^2y^3$

Worked solution.

The highest common factor (HCF) of 5 and 15 is 5. The highest common power of x is x^2 , and for y it is y^2 .

$$5x^2y^2(x - 3y)$$

Identify the highest common factors in the expression.

Final answer: $5x^2y^2(x - 3y)$

Area-model diagram.

$\times$	x	$-3y$
$5x^2y^2$	$5x^3y^2$	$-15x^2y^3$

HCF is $5x^2y^2$ (lowest power of each variable).

Factorising 3 (a2-003)

Difficulty: Foundation / Marks: 2

Factorise completely: $4a^2bc + 12ab^2c - 8abc^2$

Worked solution.

Step 1: The HCF of the numbers (4, 12, 8) is 4. The common variables are a , b , and c .

$$4abc$$

Identify the highest common numerical and algebraic factors in the expression.

Step 2: Factor out the HCF from each term in the expression.

$$4abc(a + 3b - 2c)$$

Divide each term by the common factor to find the contents of the bracket.

Final answer: $4abc(a + 3b - 2c)$

Area-model diagram.

$\times$	a	$3b$	$-2c$
$4abc$	$4a^2bc$	$12ab^2c$	$-8abc^2$

HCF $4abc$; the bracket has three terms.

Factorising 4 (a2-004)

Difficulty: Foundation / Marks: 2

Write the expression as a product of factors: $p^2 - 16q^2$

Worked solution.

Recognise that the expression is a difference of two squares, where $16q^2$ is $(4q)^2$.

$$p^2 - (4q)^2$$

Identify the squared terms.

Final answer: $(p - 4q)(p + 4q)$

Area-model diagram.

$\times$	p	$-4q$
p	p^2	$-4pq$
$4q$	$4pq$	$-16q^2$

Difference of two squares: the off-diagonal terms $-4pq$ and $+4pq$ cancel.

Factorising 5 (a2-005)

Difficulty: Foundation / Marks: 3

Factorise completely: $50x^2 - 18y^2$

Worked solution.

Step 1: Extract the common factor of 2 from both terms.

$$2(25x^2 - 9y^2)$$

Find the numerical highest common factor.

Step 2: Apply the difference of two squares to the expression inside the bracket.

$$2(5x - 3y)(5x + 3y)$$

Factorise the remaining quadratic expression.

Final answer: $2(5x - 3y)(5x + 3y)$

Area-model diagram.

$\times$	$5x$	$-3y$
$5x$	$25x^2$	$-15xy$
$3y$	$15xy$	$-9y^2$

Take out the common factor 2 first, then this grid factorises $25x^2 - 9y^2$ (DOTS).

Factorising 6 (a2-006)

Difficulty: Foundation / Marks: 2

Write the expression as a product of factors: $m^2 - 7$

Worked solution.

Treat 7 as the square of a surd, $(\sqrt{7})^2$, to apply the difference of two squares.

$$m^2 - (\sqrt{7})^2$$

Set up the difference of two squares using a square root.

Final answer: $(m - \sqrt{7})(m + \sqrt{7})$

Area-model diagram.

$\times$	m	$-\sqrt{7}$
m	m^2	$-\sqrt{7}m$
$\sqrt{7}$	$\sqrt{7}m$	-7

Treat 7 as $(\sqrt{7})^2$; the cross terms cancel.

Factorising 7 (a2-007)

Difficulty: Foundation / Marks: 2

Express as the product of factors: $(x + y)^2 + 5(x + y)$

Worked solution.

Notice that the binomial $(x + y)$ is a common factor in both parts of the expression.

$$(x + y)[(x + y) + 5]$$

Extract the common bracket.

Final answer: $(x + y)(x + y + 5)$

Area-model diagram.

$\times$	$(x + y)$	5
$(x + y)$	$(x + y)^2$	$5(x + y)$

The whole bracket $(x + y)$ is the common factor.

Factorising 8 (a2-008)

Difficulty: Foundation / Marks: 3

Express as the product of factors: $a^2(b - 3c) + 4d(3c - b)$

Worked solution.

Step 1: The brackets $(b - 3c)$ and $(3c - b)$ are opposites. Factor out -1 from the second term to make them match.

$$a^2(b - 3c) - 4d(b - 3c)$$

Reverse the sign to reveal a common bracket.

Step 2: Factor out the newly found common bracket.

$$(b - 3c)(a^2 - 4d)$$

Extract the common binomial factor.

Final answer: $(b - 3c)(a^2 - 4d)$

Area-model diagram.

$\times$	a^2	$-4d$
$(b - 3c)$	$a^2(b - 3c)$	$-4d(b - 3c)$

Rewrite $4d(3c - b) = -4d(b - 3c)$ so both terms share $(b - 3c)$.

Factorising 9 (a2-009)

Difficulty: Foundation / Marks: 3

Simplify the expression, leaving your answer in factorised form: $3(m - n)^2 - 6m(m - n)$

Worked solution.

Step 1: Identify and factor out the common factor of $3(m - n)$.

$$3(m - n)[(m - n) - 2m]$$

Extract the shared numerical and binomial factors.

Step 2: Simplify the expression inside the square brackets: $m - n - 2m = -m - n$.

$$3(m - n)(-m - n)$$

Collect like terms inside the remaining factor.

Final answer: $-3(m - n)(m + n)$

Area-model diagram.

$\times$	$(m - n)$	$-2m$
$3(m - n)$	$3(m - n)^2$	$-6m(m - n)$

Common factor $3(m - n)$; then $(m - n) - 2m = -(m + n)$ gives $-3(m - n)(m + n)$.

Factorising 10 (a2-010)

Difficulty: Foundation / Marks: 3

Factorise fully: $(p + 2q)^2 + p + 2q$

Worked solution.

Step 1: Group the last two terms together to reveal a hidden common binomial factor.

$$(p + 2q)^2 + 1(p + 2q)$$

Recognise that $p + 2q$ is identical to the expression inside the squared bracket.

Step 2: Factor out the common bracket $(p + 2q)$.

$$(p + 2q)[(p + 2q) + 1]$$

Extract the shared factor from both parts of the expression.

Final answer: $(p + 2q)(p + 2q + 1)$

Area-model diagram.

$\times$	$(p + 2q)$	1
$(p + 2q)$	$(p + 2q)^2$	$(p + 2q)$

Common bracket $(p + 2q)$; the second term contributes a $+1$.

Factorising 11 (a2-011)

Difficulty: Foundation / Marks: 3

Factorise completely: $(x + 3)^2 - 25$

Worked solution.

Step 1: Recognise this is a difference of two squares, where 25 is 5^2 and $A = (x + 3)$.

$$[(x + 3) - 5][(x + 3) + 5]$$

Apply the pattern $A^2 - B^2 = (A - B)(A + B)$.

Step 2: Simplify the expressions inside each of the new brackets.

$$(x - 2)(x + 8)$$

Collect the number terms together.

Final answer: $(x - 2)(x + 8)$

Area-model diagram.

$\times$	$(x + 3)$	-5
$(x + 3)$	$(x + 3)^2$	$-5(x + 3)$
5	$5(x + 3)$	-25

DOTS with $a = (x + 3)$, $b = 5$: $(x + 3 - 5)(x + 3 + 5) = (x - 2)(x + 8)$.

Factorising 12 (a2-012)

Difficulty: Foundation / Marks: 2

Factorise by grouping: $ac + ad + bc + bd$

Worked solution.

Step 1: Split the expression in half and factorise the first two terms and the last two terms separately.

$$a(c + d) + b(c + d)$$

Extract the common factor a from the first pair, and b from the second pair.

Step 2: Factor out the newly formed common bracket $(c + d)$.

$$(c + d)(a + b)$$

Combine the factored groups.

Final answer: $(c + d)(a + b)$

Area-model diagram.

$\times$	c	d
a	ac	ad
b	bc	bd

Grouping: the four given terms are exactly the four cells; read $(a + b)(c + d)$ off the edges.

Factorising 13 (a2-013)

Difficulty: Challenge / Marks: 3

Simplify the expression, leaving your answer in factorised form: $(l + w + h)^2 - l(l + w + h)$

Worked solution.

Step 1: Identify $(l + w + h)$ as a common factor in both terms and extract it.

$$(l + w + h)[(l + w + h) - l]$$

Pull the shared bracket out to the front.

Step 2: Simplify the expression inside the square brackets: $l + w + h - l = w + h$.

$$(l + w + h)(w + h)$$

Cancel out the l and $-l$.

Final answer: $(l + w + h)(w + h)$

Area-model diagram.

$\times$	$(l + w + h)$	$-l$
$(l + w + h)$	$(l + w + h)^2$	$-l(l + w + h)$

Common bracket $(l + w + h)$; then $(l + w + h) - l = (w + h)$.

Factorising 14 (a2-014)

Difficulty: Foundation / Marks: 2

Factorise completely: $-4x^2y - 8xy^2$

Worked solution.

Step 1: Identify the highest common factor, including the negative sign: $-4xy$.

$$-4xy$$

Both terms share a factor of -4 , x , and y .

Step 2: Divide both original terms by $-4xy$ to find the contents of the bracket.

$$-4xy(x + 2y)$$

Ensure the signs inside the bracket become positive because a negative was factored out.

Final answer: $-4xy(x + 2y)$

Area-model diagram.

$\times$	x	$2y$
$-4xy$	$-4x^2y$	$-8xy^2$

Take out $-4xy$ (negative HCF) so the bracket starts with $+x$.

Factorising 15 (a2-015)

Difficulty: Foundation / Marks: 4

Factorise fully: $16a^4 - b^4$

Worked solution.

Step 1: Treat this as a difference of two squares, where the terms are $(4a^2)^2$ and $(b^2)^2$.

$$(4a^2 - b^2)(4a^2 + b^2)$$

Apply the standard difference of two squares rule to the power of 4 terms.

Step 2: Recognise that the first bracket, $(4a^2 - b^2)$, is another difference of two squares.

$$(2a - b)(2a + b)(4a^2 + b^2)$$

Factorise the resulting quadratic, leaving the sum of squares $(4a^2 + b^2)$ as it cannot be factorised further.

Final answer: $(2a - b)(2a + b)(4a^2 + b^2)$

Area-model diagram.

$\times$	$4a^2$	$-b^2$
$4a^2$	$16a^4$	$-4a^2b^2$
b^2	$4a^2b^2$	$-b^4$

First DOTS gives $(4a^2 - b^2)(4a^2 + b^2)$; then $4a^2 - b^2 = (2a - b)(2a + b)$ is another DOTS.

Factorising 16 (a2-016)

Difficulty: Foundation / Marks: 4

Simplify the expression, leaving your answer in factorised form: $x^2(1 - y)^2 + x(2 - 2y)$

Worked solution.

Step 1: Factorise the second term, $x(2 - 2y)$, by extracting a 2 to reveal a common bracket.

$$x^2(1 - y)^2 + 2x(1 - y)$$

Make the brackets match across both terms.

Step 2: Identify and factor out the common factor of $x(1 - y)$ from the entire expression.

$$x(1 - y)[x(1 - y) + 2]$$

Extract the shared algebraic and binomial components.

Step 3: Expand the inner bracket slightly to neaten the final expression (optional but good practice).

$$x(1 - y)(x - xy + 2)$$

Distribute the x inside the square brackets.

Final answer: $x(1 - y)(x - xy + 2)$

Area-model diagram.

$\times$	$x(1 - y)$	2
$x(1 - y)$	$x^2(1 - y)^2$	$2x(1 - y)$

Since $2 - 2y = 2(1 - y)$, the common factor is $x(1 - y)$; then $x(1 - y) + 2 = x - xy + 2$.

Factorising 17 (a2-017)

Difficulty: Foundation / Marks: 1

Factorise: $15x + 20y$

Worked solution.

Find the highest common factor (HCF) of the numbers 15 and 20, which is 5.

$$5(3x + 4y)$$

Divide both terms by 5 and place the remainder inside the bracket.

Final answer: $5(3x + 4y)$

Area-model diagram.

$\times$	$3x$	$4y$
5	$15x$	$20y$

HCF 5.

Factorising 18 (a2-018)

Difficulty: Standard / Marks: 2

Factorise fully: $6a^2b - 9ab^2$

Worked solution.

Step 1: Identify the HCF of 6 and 9 (which is 3) and the common variables (which are a and b).

$$3ab$$

Find the highest algebraic and numerical factors shared by both terms.

Step 2: Factor out 3ab from the original expression.

$$3ab(2a - 3b)$$

Divide each term by $3ab$ to find the expression inside the bracket.

Final answer: $3ab(2a - 3b)$

Area-model diagram.

$\times$	$2a$	$-3b$
$3ab$	$6a^2b$	$-9ab^2$

HCF $3ab$.

Factorising 19 (a2-019)

Difficulty: Foundation / Marks: 1

Write the expression as a product of factors: $y^2 - 81$

Worked solution.

Recognise that this is a difference of two squares, where 81 is 9^2 .

$$y^2 - 9^2$$

Identify the squared terms to apply the rule $A^2 - B^2 = (A - B)(A + B)$.

Final answer: $(y - 9)(y + 9)$

Area-model diagram.

$\times$	y	-9
y	y^2	$-9y$
9	$9y$	-81

DOTS: $81 = 9^2$, cross terms cancel.

Factorising 20 (a2-020)

Difficulty: Foundation / Marks: 3

Factorise completely: $3x^2 - 75$

Worked solution.

Step 1: Extract the common numerical factor of 3 from both terms first.

$$3(x^2 - 25)$$

Always check for a common factor before applying other factorisation rules.

Step 2: Factorise the expression inside the bracket using the difference of two squares rule.

$$3(x - 5)(x + 5)$$

Recognise that 25 is 5^2 .

Final answer: $3(x - 5)(x + 5)$

Area-model diagram.

×	x	-5
x	x^2	$-5x$
5	$5x$	-25

Take out 3 first: $3(x^2 - 25)$, then DOTS on $x^2 - 25$.

Factorising 21 (a2-021)

Difficulty: Foundation / Marks: 2

Factorise by grouping: $xy + 2x + 3y + 6$

Worked solution.

Step 1: Group the first two terms and the last two terms, then factorise them separately.

$$x(y + 2) + 3(y + 2)$$

Extract x from the first pair and 3 from the second pair.

Step 2: Factor out the common binomial bracket $(y + 2)$.

$$(y + 2)(x + 3)$$

Combine the factored groups together.

Final answer: $(y + 2)(x + 3)$

Area-model diagram.

×	y	2
x	xy	$2x$
3	$3y$	6

Grouping: $x(y + 2) + 3(y + 2) = (x + 3)(y + 2)$.

Factorising 22 (a2-022)

Difficulty: Foundation / Marks: 2

Express as a product of factors: $(a - b)^2 + 2(a - b)$

Worked solution.

Step 1: Identify that the bracket $(a - b)$ is a common factor in both terms.

$$(a - b)[(a - b) + 2]$$

Extract the shared binomial bracket to the front.

Step 2: Remove the inner square brackets for the final simplified answer.

$$(a - b)(a - b + 2)$$

Neaten up the expression.

Final answer: $(a - b)(a - b + 2)$

Area-model diagram.

$\times$	$(a - b)$	2
$(a - b)$	$(a - b)^2$	$2(a - b)$

Common bracket $(a - b)$.

Factorising 23 (a2-023)

Difficulty: Foundation / Marks: 3

Factorise fully: $4x^3y - 16xy^3$

Worked solution.

Step 1: Find and extract the highest common factor for both terms, which is $4xy$.

$$4xy(x^2 - 4y^2)$$

Divide both terms by $4xy$ to find the remaining expression.

Step 2: Recognise that the expression inside the bracket is a difference of two squares and factorise it.

$$4xy(x - 2y)(x + 2y)$$

Apply the pattern $A^2 - B^2 = (A - B)(A + B)$ to $(x^2 - 4y^2)$.

Final answer: $4xy(x - 2y)(x + 2y)$

Area-model diagram.

$\times$	x	$-2y$
x	x^2	$-2xy$
$2y$	$2xy$	$-4y^2$

Take out $4xy$ first: $4xy(x^2 - 4y^2)$, then DOTS.

Factorising 24 (a2-024)

Difficulty: Foundation / Marks: 1

Write the expression as a product of factors: $100 - z^2$

Worked solution.

Recognise that this is a difference of two squares, where 100 is 10^2 .

$$10^2 - z^2$$

Identify the squared terms to apply the rule $A^2 - B^2 = (A - B)(A + B)$.

Final answer: $(10 - z)(10 + z)$

Area-model diagram.

$\times$	10	$-z$
10	100	$-10z$
z	$10z$	$-z^2$

DOTS: $100 = 10^2$.

Factorising 25 (a2-025)

Difficulty: Foundation / Marks: 2

Factorise completely: $14m^2n^3 + 21m^3n^2$

Worked solution.

Step 1: Identify the highest common factor of the numbers (which is 7) and the highest common powers of the variables (which are m^2 and n^2).

$$7m^2n^2$$

Find the greatest term that divides exactly into both parts of the expression.

Step 2: Factor out $7m^2n^2$ from the original expression.

$$7m^2n^2(2n + 3m)$$

Divide each term by the common factor to find the contents of the bracket.

Final answer: $7m^2n^2(2n + 3m)$

Area-model diagram.

$\times$	$2n$	$3m$
$7m^2n^2$	$14m^2n^3$	$21m^3n^2$

HCF $7m^2n^2$.

Factorising 26 (a2-026)

Difficulty: Foundation / Marks: 2

Express as the product of factors: $x(x - 4) - 3(x - 4)$

Worked solution.

Identify that the bracket $(x - 4)$ is a common factor in both terms.

$$(x - 4)(x - 3)$$

Extract the shared binomial bracket to the front, placing the remaining outer terms into a second bracket.

Final answer: $(x - 4)(x - 3)$

Area-model diagram.

$\times$	x	-3
$(x - 4)$	$x(x - 4)$	$-3(x - 4)$

Common bracket $(x - 4)$; remaining factor $(x - 3)$.

Factorising 27 (a2-027)

Difficulty: Foundation / Marks: 3

Factorise fully: $8x^2 - 32y^2$

Worked solution.

Step 1: Extract the common numerical factor of 8 from both terms first.

$$8(x^2 - 4y^2)$$

Always check for a highest common factor before looking for other patterns.

Step 2: Recognise that the expression inside the bracket is a difference of two squares and factorise it.

$$8(x - 2y)(x + 2y)$$

Apply the pattern $A^2 - B^2 = (A - B)(A + B)$ to the inner quadratic.

Final answer: $8(x - 2y)(x + 2y)$

Area-model diagram.

$\times$	x	$-2y$
x	x^2	$-2xy$
$2y$	$2xy$	$-4y^2$

Take out 8 first: $8(x^2 - 4y^2)$, then DOTS.

Factorising 28 (a2-028)

Difficulty: Foundation / Marks: 2

Factorise by grouping: $ax - bx + ay - by$

Worked solution.

Step 1: Group the first two terms and the last two terms, factorising each pair separately.

$$x(a - b) + y(a - b)$$

Extract x from the first pair and y from the second pair.

Step 2: Factor out the newly formed common binomial bracket $(a - b)$.

$$(a - b)(x + y)$$

Combine the factored groups to leave a product of two brackets.

Final answer: $(a - b)(x + y)$

Area-model diagram.

$\times$	a	$-b$
x	ax	$-bx$
y	ay	$-by$

Grouping: $x(a - b) + y(a - b) = (x + y)(a - b)$.

Factorising 29 (a2-029)

Difficulty: Foundation / Marks: 3

Express as the product of factors: $2u(v - w) + 5(w - v)$

Worked solution.

Step 1: Notice the brackets $(v - w)$ and $(w - v)$ are opposites. Factor out -1 from the second term to make them identical.

$$2u(v - w) - 5(v - w)$$

Reverse the sign of the second term to reveal a common bracket.

Step 2: Factor out the common bracket $(v - w)$.

$$(v - w)(2u - 5)$$

Extract the shared binomial factor.

Final answer: $(v - w)(2u - 5)$

Area-model diagram.

$\times$	$2u$	-5
$(v - w)$	$2u(v - w)$	$-5(v - w)$

Rewrite $5(w - v) = -5(v - w)$ to expose the common bracket $(v - w)$.

Factorising 30 (a2-030)

Difficulty: Foundation / Marks: 2

Simplify the expression, leaving your answer in factorised form: $(p + q)^2 - 3(p + q)$

Worked solution.

Step 1: Identify that $(p + q)$ is a common factor in both terms.

$$(p + q)[(p + q) - 3]$$

Pull the shared bracket out to the front.

Step 2: Remove the inner square brackets to give the final simplified expression.

$$(p + q)(p + q - 3)$$

Neaten up the expression.

Final answer: $(p + q)(p + q - 3)$

Area-model diagram.

$\times$	$(p + q)$	-3
$(p + q)$	$(p + q)^2$	$-3(p + q)$

Common bracket $(p + q)$.

Factorising 31 (a2-031)

Difficulty: Foundation / Marks: 1

Factorise: $18p - 27q$

Worked solution.

Find the highest common factor (HCF) of 18 and 27, which is 9.

$$9(2p - 3q)$$

Divide both terms by 9 to find the expression inside the bracket.

Final answer: $9(2p - 3q)$

Area-model diagram.

×	$2p$	$-3q$
9	$18p$	$-27q$

HCF 9.

Factorising 32 (a2-032)

Difficulty: Foundation / Marks: 2

Write the expression as a product of factors: $4c^2 - d^2$

Worked solution.

Recognise this as a difference of two squares, rewriting the terms as $(2c)^2$ and d^2 .

$$(2c)^2 - d^2$$

Identify the squared components to apply the rule $A^2 - B^2 = (A - B)(A + B)$.

Final answer: $(2c - d)(2c + d)$

Area-model diagram.

×	$2c$	$-d$
$2c$	$4c^2$	$-2cd$
d	$2cd$	$-d^2$

DOTS: $4c^2 = (2c)^2$.

Factorising 33 (a2-033)

Difficulty: Foundation / Marks: 2

Factorise fully: $5x^2y + 15xy^2 - 10xy$

Worked solution.

Step 1: Identify the highest common numerical factor (which is 5) and the shared variables (which are x and y).

$$5xy$$

Find the greatest term that divides exactly into all three parts of the expression.

Step 2: Factor out $5xy$ from the original expression.

$$5xy(x + 3y - 2)$$

Divide each original term by the common factor to populate the bracket.

Final answer: $5xy(x + 3y - 2)$

Area-model diagram.

$\times$	x	$3y$	-2
$5xy$	$5x^2y$	$15xy^2$	$-10xy$

HCF $5xy$; three-term bracket.

Factorising 34 (a2-034)

Difficulty: Foundation / Marks: 2

Express as the product of factors: $m(n - 2) + 4(n - 2)$

Worked solution.

Identify that the binomial $(n - 2)$ is a common factor in both terms.

$$(n - 2)(m + 4)$$

Extract the shared bracket to the front and group the remaining outer terms into a second bracket.

Final answer: $(n - 2)(m + 4)$

Area-model diagram.

$\times$	m	4
$(n - 2)$	$m(n - 2)$	$4(n - 2)$

Common bracket $(n - 2)$.

Factorising 35 (a2-035)

Difficulty: Foundation / Marks: 3

Factorise completely: $12a^2 - 75b^2$

Worked solution.

Step 1: Extract the highest common numerical factor of 3 from both terms first.

$$3(4a^2 - 25b^2)$$

Always check for a common factor before applying difference of two squares.

Step 2: Recognise that the expression inside the bracket is a difference of two squares and factorise it.

$$3(2a - 5b)(2a + 5b)$$

Apply the pattern $A^2 - B^2 = (A - B)(A + B)$ to the inner quadratic.

Final answer: $3(2a - 5b)(2a + 5b)$

Area-model diagram.

×	$2a$	$-5b$
$2a$	$4a^2$	$-10ab$
$5b$	$10ab$	$-25b^2$

Take out 3 first: $3(4a^2 - 25b^2)$, then DOTS.

Factorising 36 (a2-036)

Difficulty: Foundation / Marks: 2

Factorise by grouping: $pr + ps + qr + qs$

Worked solution.

Step 1: Group the first two terms and the last two terms, then factorise each pair.

$$p(r + s) + q(r + s)$$

Extract p from the first pair and q from the second pair.

Step 2: Factor out the newly formed common binomial bracket $(r + s)$.

$$(r + s)(p + q)$$

Combine the factored groups together.

Final answer: $(r + s)(p + q)$

Area-model diagram.

$\times$	r	s
p	pr	ps
q	qr	qs

Grouping: $p(r + s) + q(r + s) = (p + q)(r + s)$.

Factorising 37 (a2-037)

Difficulty: Foundation / Marks: 3

Simplify the expression, leaving your answer in factorised form: $(2x - y)^2 - x(2x - y)$

Worked solution.

Step 1: Identify that $(2x - y)$ is a common factor in both terms and extract it.

$$(2x - y)[(2x - y) - x]$$

Pull the shared binomial bracket out to the front.

Step 2: Simplify the expression inside the square brackets by collecting like terms: $2x - y - x = x - y$.

$$(2x - y)(x - y)$$

Neaten up the second bracket to find the final factorised form.

Final answer: $(2x - y)(x - y)$

Area-model diagram.

$\times$	$(2x - y)$	$-x$
$(2x - y)$	$(2x - y)^2$	$-x(2x - y)$

Common bracket $(2x - y)$; then $(2x - y) - x = (x - y)$.

Factorising 38 (a2-038)

Difficulty: Foundation / Marks: 1

Factorise completely: $24u + 36v$

Worked solution.

Identify the highest common factor (HCF) of the numbers 24 and 36, which is 12.

$$12(2u + 3v)$$

Divide both terms by 12 and place the remaining expression inside the bracket.

Final answer: $12(2u + 3v)$

Area-model diagram.

$\times$	$2u$	$3v$
12	$24u$	$36v$

HCF 12.

Factorising 39 (a2-039)

Difficulty: Foundation / Marks: 2

Write the expression as a product of factors: $49x^2 - y^2$

Worked solution.

Recognise that this is a difference of two squares, rewriting the first term as $(7x)^2$.

$$(7x)^2 - y^2$$

Identify the squared components to apply the rule $A^2 - B^2 = (A - B)(A + B)$.

Final answer: $(7x - y)(7x + y)$

Area-model diagram.

$\times$	$7x$	$-y$
$7x$	$49x^2$	$-7xy$
y	$7xy$	$-y^2$

DOTS: $49x^2 = (7x)^2$.

Factorising 40 (a2-040)

Difficulty: Foundation / Marks: 2

Factorise fully: $16a^3b^2 - 24a^2b^3$

Worked solution.

Step 1: Identify the highest common numerical factor (which is 8) and the highest common powers of the variables (which are a^2 and b^2).

$$8a^2b^2$$

Find the greatest algebraic term that divides exactly into both parts of the expression.

Step 2: Factor out $8a^2b^2$ from the original expression.

$$8a^2b^2(2a - 3b)$$

Divide each original term by the common factor to populate the bracket.

Final answer: $8a^2b^2(2a - 3b)$

Area-model diagram.

$\times$	$2a$	$-3b$
$8a^2b^2$	$16a^3b^2$	$-24a^2b^3$

HCF $8a^2b^2$.

Factorising 41 (a2-041)

Difficulty: Foundation / Marks: 2

Express as the product of factors: $x(y + 5) - 2(y + 5)$

Worked solution.

Identify that the binomial $(y + 5)$ is a common factor in both terms.

$$(y + 5)(x - 2)$$

Extract the shared bracket to the front and group the remaining outer multipliers into a second bracket.

Final answer: $(y + 5)(x - 2)$

Area-model diagram.

$\times$	x	-2
$(y + 5)$	$x(y + 5)$	$-2(y + 5)$

Common bracket $(y + 5)$.

Factorising 42 (a2-042)

Difficulty: Foundation / Marks: 3

Factorise completely: $5m^3 - 45m$

Worked solution.

Step 1: Extract the highest common algebraic and numerical factor, $5m$, from both terms first.

$$5m(m^2 - 9)$$

Always check for a common factor before checking for quadratics or difference of two squares.

Step 2: Recognise that the expression inside the bracket is a difference of two squares and factorise it.

$$5m(m - 3)(m + 3)$$

Apply the pattern $A^2 - B^2 = (A - B)(A + B)$ to the inner expression.

Final answer: $5m(m - 3)(m + 3)$

Area-model diagram.

$\times$	m	-3
m	m^2	$-3m$
3	$3m$	-9

Take out $5m$ first: $5m(m^2 - 9)$, then DOTS.

Factorising 43 (a2-043)

Difficulty: Foundation / Marks: 2

Factorise by grouping: $cx + dx - cy - dy$

Worked solution.

Step 1: Group the first two terms and the last two terms, then factorise each pair. Be careful with the negative signs in the second pair.

$$x(c + d) - y(c + d)$$

Extract x from the first pair and $-y$ from the second pair to ensure the inner brackets match.

Step 2: Factor out the newly formed common binomial bracket $(c + d)$.

$$(c + d)(x - y)$$

Combine the factored groups together.

Final answer: $(c + d)(x - y)$

Area-model diagram.

$\times$	c	d
x	cx	dx
$-y$	$-cy$	$-dy$

Grouping: $x(c + d) - y(c + d) = (x - y)(c + d)$.

Factorising 44 (a2-044)

Difficulty: Foundation / Marks: 3

Simplify the expression, leaving your answer in factorised form: $(2x + 3)^2 - (2x + 3)(x - 1)$

Worked solution.

Step 1: Identify that $(2x + 3)$ is a common factor in both terms and extract it.

$$(2x + 3)[(2x + 3) - (x - 1)]$$

Pull the shared binomial bracket out to the front.

Step 2: Carefully expand the inner expression, paying attention to the negative sign in front of the $(x - 1)$ bracket.

$$(2x + 3)(2x + 3 - x + 1)$$

Distribute the negative sign to the terms inside the second bracket.

Step 3: Simplify the expression inside the second bracket by collecting like terms.

$$(2x + 3)(x + 4)$$

Combine the x terms and the number terms.

Final answer: $(2x + 3)(x + 4)$

Area-model diagram.

$\times$	$(2x + 3)$	$-(x - 1)$
$(2x + 3)$	$(2x + 3)^2$	$-(2x + 3)(x - 1)$

Common bracket $(2x + 3)$; then $(2x + 3) - (x - 1) = (x + 4)$.

Factorising 45 (a2-045)

Difficulty: Foundation / Marks: 1

Factorise: $12x - 18$

Worked solution.

Find the highest common factor (HCF) of 12 and 18, which is 6.

$$6(2x - 3)$$

Divide both terms by 6 to find the expression inside the bracket.

Final answer: $6(2x - 3)$

Area-model diagram.

$\times$	$2x$	-3
6	$12x$	-18

HCF 6.

Factorising 46 (a2-046)

Difficulty: Foundation / Marks: 2

Factorise fully: $5a^2b + 10ab^2$

Worked solution.

Step 1: Identify the highest common numerical factor (which is 5) and the shared variables (which are a and b).

$$5ab$$

Find the greatest term that divides exactly into both parts of the expression.

Step 2: Factor out 5ab from the original expression.

$$5ab(a + 2b)$$

Divide each original term by the common factor to populate the bracket.

Final answer: $5ab(a + 2b)$

Area-model diagram.

$\times$	a	$2b$
$5ab$	$5a^2b$	$10ab^2$

HCF 5ab.

Factorising 47 (a2-047)

Difficulty: Foundation / Marks: 2

Write the expression as a product of factors: $81p^2 - 16q^2$

Worked solution.

Recognise this as a difference of two squares, rewriting the terms as $(9p)^2$ and $(4q)^2$.

$$(9p)^2 - (4q)^2$$

Identify the squared components to apply the rule $A^2 - B^2 = (A - B)(A + B)$.

Final answer: $(9p - 4q)(9p + 4q)$

Area-model diagram.

$\times$	$9p$	$-4q$
$9p$	$81p^2$	$-36pq$
$4q$	$36pq$	$-16q^2$

DOTS: $81p^2 = (9p)^2$, $16q^2 = (4q)^2$.

Factorising 48 (a2-048)

Difficulty: Foundation / Marks: 2

Express as the product of factors: $2x(y - 1) + 3(y - 1)$

Worked solution.

Identify that the binomial $(y - 1)$ is a common factor in both terms.

$$(y - 1)(2x + 3)$$

Extract the shared bracket to the front and group the remaining outer terms into a second bracket.

Final answer: $(y - 1)(2x + 3)$

Area-model diagram.

$\times$	$2x$	3
$(y - 1)$	$2x(y - 1)$	$3(y - 1)$

Common bracket $(y - 1)$.

Factorising 49 (a2-049)

Difficulty: Foundation / Marks: 3

Express as the product of factors: $m(n - 3) - 2(3 - n)$

Worked solution.

Step 1: Notice the brackets $(n - 3)$ and $(3 - n)$ are opposites. Factor out -1 from the second term to make them identical.

$$m(n - 3) + 2(n - 3)$$

Reverse the sign of the second term to reveal a common bracket.

Step 2: Factor out the common bracket $(n - 3)$.

$$(n - 3)(m + 2)$$

Extract the shared binomial factor.

Final answer: $(n - 3)(m + 2)$

Area-model diagram.

$\times$	m	2
$(n - 3)$	$m(n - 3)$	$2(n - 3)$

Rewrite $-2(3 - n) = +2(n - 3)$ to expose the common bracket $(n - 3)$.

Factorising 50 (a2-050)

Difficulty: Foundation / Marks: 3

Factorise completely: $18x^2 - 50$

Worked solution.

Step 1: Extract the highest common numerical factor of 2 from both terms first.

$$2(9x^2 - 25)$$

Always check for a common factor before applying difference of two squares.

Step 2: Recognise that the expression inside the bracket is a difference of two squares and factorise it.

$$2(3x - 5)(3x + 5)$$

Apply the pattern $A^2 - B^2 = (A - B)(A + B)$ to the inner quadratic.

Final answer: $2(3x - 5)(3x + 5)$

Area-model diagram.

$\times$	$3x$	-5
$3x$	$9x^2$	$-15x$
5	$15x$	-25

Take out 2 first: $2(9x^2 - 25)$, then DOTS.
